

存储性能测试(fio)

一、工具简介：

fio是一个开源的I/O压力测试工具，主要是用来测试磁盘/SSD的IO性能，即测试磁盘IOPS、带宽等。

二、测试目的：

测试存储的读写性能，包含：随机读、随机写、顺序读、顺序写、混合随机读写。

三、使用方法：

安装后即可使用。

四、工具下载：

官网下载地址：<http://freshmeat.sourceforge.net/projects/fio>

deb包下载地址：<http://archive.ubuntu.com/ubuntu/pool/universe/f/fio/>

五、工具安装：

- 1、apt安装：apt install fio
- 2、yum安装：yum install fio
- 3、源码包编译安装：
 - 1)、解压工具包：unzip fio-master.zip
 - 2)、解压后，进入到工具目录中：cd fio-master
 - 3)、编译：./configure
 - 4)、编译：make
 - 5)、安装：make install

六、测试场景举例：

- 1、在/root目录下创建一个名为“test4G”、4GB大小的测试文件：

```
dd if=/dev/zero of=/root/test4G bs=4k count=1048576
```

```
root@hwos:~/fio/fio-master# dd if=/dev/zero of=/root/test4G bs=4k count=1048576
1048576+0 records in
1048576+0 records out
4294967296 bytes (4.3 GB, 4.0 GiB) copied, 10.7741 s, 399 MB/s
root@hwos:~/fio/fio-master#
root@hwos:~/fio/fio-master#
```

2、随机读测试：

fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=randread -ioengine=psync -
bs=4k -numjobs=64 -runtime=300 -group_reporting -name=randread.log

```
root@hwos:~/fio/fio-master# fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=randread -ioengine=psync -bs=4k -numjobs=64 -runtime=300 -group_reporting -name=randread.log
randread.log: (g=0): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
...
fio-3.15
Starting 64 threads
Jobs: 64 (f=64): [r(64)][100.0%][r=232MiB/s][r=59.5k IOPS][eta 00m:00s]
randread.log: (groupid=0, jobs=64): err= 0: pid=3493: Thu Mar 19 17:16:58 2020
read: IOPS=58.9k, BW=230MiB/s (241MB/s) (67.4GiB/300013msec)
clat (usec): min=43, max=109199, avg=1073.39, stdev=4043.75
lat (usec): min=44, max=109200, avg=1074.59, stdev=4044.13
clat percentiles (usec):
| 1.00th=[ 66], 5.00th=[ 76], 10.00th=[ 81], 20.00th=[ 90],
| 30.00th=[ 97], 40.00th=[ 105], 50.00th=[ 111], 60.00th=[ 118],
| 70.00th=[ 133], 80.00th=[ 165], 90.00th=[ 289], 95.00th=[11731],
| 99.00th=[21103], 99.50th=[23725], 99.90th=[28443], 99.95th=[30278],
| 99.99th=[34866]
bw ( KiB/s): min=68508, max=342494, per=100.00%, avg=236361.38, stdev=538.46, samples=38209
iops: min=17126, max=85623, avg=59088.98, stdev=134.61, samples=38209
lat (usec): 50=0.02%, 100=34.50%, 250=54.32%, 500=3.75%, 750=0.96%
lat (msec): 1000=0.36%
lat (msec): 2=0.46%, 4=0.24%, 10=0.17%, 20=3.85%, 50=1.37%
lat (msec): 100=0.01%, 250=0.01%
cpu: usr=0.43%, sys=5.08%, ctx=24664818, majf=0, minf=73
IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=17676461,0,0,0 short=0,0,0,0 dropped=0,0,0,0
latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
READ: bw=230MiB/s (241MB/s), 230MiB/s-230MiB/s (241MB/s-241MB/s), io=67.4GiB (72.4GB), run=300013-300013msec

Disk stats (read/write):
sda: ios=1766344/211, merge=0/210, ticks=1102960/240, in_queue=1092970, util=99.6%
root@hwos:~/fio/fio-master#
root@hwos:~/fio/fio-master#
```

运行命令

测试结果说明:
fio的结果报告内容非常丰富,但我们主要关注下面两项:
1. IOPS, 即磁盘的每秒读写次数,这个是随机读写考察的重点。IOPS值越高表示磁盘读写数据越快。IOPS不能设置,是出厂就已经定好的,所以IOPS是衡量磁盘读写性能的重要指标之一。
2. BW, 即磁盘的吞吐量,这个是顺序读写考察的重点。

3、随机写测试：

fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=randwrite -ioengine=psync -
bs=4k -numjobs=64 -runtime=300 -group_reporting -name=randwrite.log

```
root@hwos:~/fio/fio-master# fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=randwrite -ioengine=psync -bs=4k -numjobs=64 -runtime=300 -group_reporting -name=randwrite.log
randwrite.log: (g=0): rw=randwrite, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
...
fio-3.15
Starting 64 threads
Jobs: 64 (f=64): [w(64)][100.0%][w=49.6MiB/s][w=12.7k IOPS][eta 00m:00s]
randwrite.log: (groupid=0, jobs=64): err= 0: pid=3655: Thu Mar 19 17:25:49 2020
write: IOPS=10.1k, BW=39.6MiB/s (41.5MB/s) (11.6GiB/300004msec)
clat (usec): min=51, max=254575, avg=6306.95, stdev=17626.97
lat (usec): min=52, max=254576, avg=6307.71, stdev=17626.97
clat percentiles (msec):
| 1.00th=[ 4], 5.00th=[ 4], 10.00th=[ 4], 20.00th=[ 4],
| 30.00th=[ 4], 40.00th=[ 4], 50.00th=[ 4], 60.00th=[ 4],
| 70.00th=[ 4], 80.00th=[ 4], 90.00th=[ 4], 95.00th=[ 4],
| 99.00th=[113], 99.50th=[116], 99.90th=[161], 99.95th=[163],
| 99.99th=[236]
bw ( KiB/s): min=14884, max=74892, per=100.00%, avg=40557.93, stdev=229.48, samples=38383
iops: min= 3718, max=18722, avg=10137.53, stdev=57.38, samples=38383
lat (usec): 100=0.13%, 250=0.01%, 500=0.01%, 750=0.01%, 1000=0.01%
lat (msec): 2=0.01%, 4=96.44%, 10=0.86%, 20=0.28%, 50=0.04%
lat (msec): 100=0.01%, 250=2.26%, 500=0.01%
cpu: usr=0.08%, sys=0.73%, ctx=6083579, majf=0, minf=0
IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=0,3042012,0,0 short=0,0,0,0 dropped=0,0,0,0
latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=39.6MiB/s (41.5MB/s), 39.6MiB/s-39.6MiB/s (41.5MB/s-41.5MB/s), io=11.6GiB (12.5GB), run=300004-300004msec

Disk stats (read/write):
sda: ios=0/3041453, merge=0/241, ticks=0/240520, in_queue=238320, util=78.36%
root@hwos:~/fio/fio-master#
root@hwos:~/fio/fio-master#
```

运行命令

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1. IOPS, 即磁盘的每秒读写次数,这个是随机读写考察的重点。IOPS值越高表示磁盘读写数据越快。IOPS不能设置,是出厂就已经定好的,所以IOPS是衡量磁盘读写性能的重要指标之一。
2. BW, 即磁盘的吞吐量,这个是顺序读写考察的重点。

4、顺序读测试：

fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=read -ioengine=psync -bs=4k -numjobs=64 -runtime=300 -group_reporting -name=read.log

```
root@hwos:~/fio/fio-master# fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=read -ioengine=psync -bs=4k -numjobs=64 -runtime=300 -group_reporting -name=read.log
read.log: (g=0): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
...
fio-3.15
Starting 64 threads
Jobs: 64 (f=64): [R(64)][100.0%][r=82.1MiB/s][r=21.0k IOPS][eta 00m:00s]
read.log: (groupid=0, jobs=64): err= 0: oid=3864: Thu Mar 19 17:41:07 2020
read: IOPS=20.7k, BW=80.8MiB/s (84.7MB/s)(23.7GiB/300005msec)
clat (usec): min=132, max=3196.2k, avg=3090.73, stdev=27580.90
lat (usec): min=133, max=3196.2k, avg=3091.47, stdev=27580.90
clat percentiles (usec):
| 1.00th=[ 619], 5.00th=[ 1029], 10.00th=[ 1254],
| 20.00th=[ 1467], 30.00th=[ 1598], 40.00th=[ 1680],
| 50.00th=[ 1778], 60.00th=[ 1876], 70.00th=[ 1991],
| 80.00th=[ 2147], 90.00th=[ 2540], 95.00th=[ 3294],
| 99.00th=[ 8979], 99.50th=[ 23725], 99.90th=[ 375391],
| 99.95th=[ 566232], 99.99th=[1115685]
bw ( KiB/s): min= 524, max=164026, per=100.00%, avg=89305.15, stdev=731.23, samples=35499
iops : min= 122, max=41006, avg=22324.77, stdev=182.81, samples=35499
lat (usec): 250=0.01%, 500=0.56%, 750=1.32%, 1000=2.50%
lat (msec): 2=66.40%, 4=26.01%, 10=2.30%, 20=0.36%, 50=0.19%
lat (msec): 100=0.10%, 250=0.11%, 500=0.08%, 750=0.04%, 1000=0.02%
lat (msec): 2000=0.01%, >=2000=0.01%
cpu : usr=0.16%, sys=0.77%, ctx=6429723, majf=0, minf=64
IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=6203782,0,0 short=0,0,0 dropped=0,0,0
latency : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
READ: bw=80.8MiB/s (84.7MB/s), 80.8MiB/s-80.8MiB/s (84.7MB/s-84.7MB/s), io=23.7GiB (25.4GB), run=300005-300005msec

Disk stats (read/write):
sda: ios=6008193/193, merge=2/200, ticks=14510060/501500, in_queue=15036590, util=100.00%
root@hwos:~/fio/fio-master#
root@hwos:~/fio/fio-master#
```

运行命令

测试结果说明:
fio的结果报告内容丰富,但我们主要关注下面两项:
1. IOPS, 即磁盘的每秒读写次数,这个是随机读写考察的重点。IOPS值越高表示磁盘读写数据越快。IOPS不能设置,是出厂就已经定了,所以IOPS是衡量磁盘读写性能的重要指标之一。
2. BW, 即磁盘的吞吐量,这个是顺序读写考察的重点。

5、顺序写测试：

fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=write -ioengine=psync -bs=4k -numjobs=64 -runtime=300 -group_reporting -name=write.log

```
root@hwos:~/fio/fio-master# fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=write -ioengine=psync -bs=4k -numjobs=64 -runtime=300 -group_reporting -name=write.log
write.log: (g=0): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
...
fio-3.15
Starting 64 threads
Jobs: 64 (f=64): [W(64)][100.0%][w=69.2MiB/s][w=17.7k IOPS][eta 00m:00s]
write.log: (groupid=0, jobs=64): err= 0: oid=13106: Fri Mar 20 09:17:03 2020
write: IOPS=13.8k, BW=53.9MiB/s (56.5MB/s)(15.8GiB/300004msec)
clat (usec): min=50, max=884855, avg=4632.41, stdev=13613.66
lat (usec): min=51, max=884855, avg=4633.17, stdev=13613.66
clat percentiles (msec):
| 1.00th=[ 4], 5.00th=[ 4], 10.00th=[ 4], 20.00th=[ 4],
| 30.00th=[ 4], 40.00th=[ 4], 50.00th=[ 4], 60.00th=[ 4],
| 70.00th=[ 4], 80.00th=[ 4], 90.00th=[ 5], 95.00th=[ 6],
| 99.00th=[ 39], 99.50th=[ 39], 99.90th=[ 153], 99.95th=[ 197],
| 99.99th=[ 751]
bw ( KiB/s): min= 506, max=82968, per=100.00%, avg=55964.86, stdev=385.11, samples=37856
iops : min= 122, max=20742, avg=13990.03, stdev=96.28, samples=37856
lat (usec): 100=0.13%, 250=0.01%, 500=0.01%, 750=0.01%, 1000=0.01%
lat (msec): 2=0.01%, 4=89.86%, 10=7.50%, 20=0.98%, 50=1.39%
lat (msec): 100=0.03%, 250=0.08%, 500=0.01%, 750=0.01%, 1000=0.01%
cpu : usr=0.11%, sys=0.89%, ctx=8280447, majf=0, minf=0
IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=0,4140786,0,0 short=0,0,0 dropped=0,0,0
latency : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=53.9MiB/s (56.5MB/s), 53.9MiB/s-53.9MiB/s (56.5MB/s-56.5MB/s), io=15.8GiB (16.0GB), run=300004-300004msec

Disk stats (read/write):
sda: ios=0/4138977, merge=0/236, ticks=0/217290, in_queue=214660, util=70.79%
root@hwos:~/fio/fio-master#
root@hwos:~/fio/fio-master#
```

运行命令

测试结果说明:
fio的结果报告内容丰富,但我们主要关注下面两项:
1. IOPS, 即磁盘的每秒读写次数,这个是随机读写考察的重点。IOPS值越高表示磁盘读写数据越快。IOPS不能设置,是出厂就已经定了,所以IOPS是衡量磁盘读写性能的重要指标之一。
2. BW, 即磁盘的吞吐量,这个是顺序读写考察的重点。

6、混合随机读写测试：

fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=randrw -rwmixread=30 -ioengine=psync -bs=4k -numjobs=64 -runtime=10 -group_reporting -name=randrwmixed.log

```
root@hwos:~/fio/fio-master# fio -filename=/root/test4G -direct=1 -iodepth 1 -thread -rw=randrw -rwmixread=30 -ioengine=psync -bs=4k -numjobs=64 -runtime=10 -group_reporting -name=randrwmixed.1
randrwmixed.log: (g=0): rw=randrw, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
...
fio-3.15
Starting 64 threads
Jobs: 64 (f=64): [m(64)][100.0%][r=10.8MiB/s,w=24.6MiB/s][r=2754,w=6295 IOPS][eta 00m:00s]
randrwmixed.log: (groupid=0, jobs=64): err=0: uid=13312: Fri Mar 20 09:20:13 2020
read: IOPS=2687, BW=10.5MiB/s (11.0MB/s)(105MiB/1000msec)
  clat (usec): min=123, max=35025, avg=7389.42, stdev=4436.78
  lat (usec): min=124, max=35026, avg=7309.11, stdev=4436.77
  clat percentiles (usec):
    | 1.00th=[ 163], 5.00th=[ 269], 10.00th=[ 429], 20.00th=[ 570],
    | 30.00th=[ 7635], 40.00th=[ 8455], 50.00th=[ 8979], 60.00th=[ 9372],
    | 70.00th=[ 9765], 80.00th=[10290], 90.00th=[10945], 95.00th=[11600],
    | 99.00th=[13435], 99.50th=[25035], 99.90th=[28967], 99.95th=[33424],
    | 99.99th=[34866]
  bw ( KIB/s): min=7285, max=14324, per=100.00%, avg=10749.11, stdev=29.99, samples=1272
  iops : min=1819, max= 3578, avg=2684.71, stdev= 7.49, samples=1272
write: IOPS=6254, BW=24.4MiB/s (25.6MB/s)(245MiB/1000msec)
  clat (usec): min=50, max=35311, avg=7081.68, stdev=4388.92
  lat (usec): min=51, max=35312, avg=7082.47, stdev=4388.91
  clat percentiles (usec):
    | 1.00th=[ 53], 5.00th=[ 174], 10.00th=[ 196], 20.00th=[ 457],
    | 30.00th=[ 7373], 40.00th=[ 8225], 50.00th=[ 8717], 60.00th=[ 9110],
    | 70.00th=[ 9503], 80.00th=[10028], 90.00th=[10683], 95.00th=[11338],
    | 99.00th=[13173], 99.50th=[24773], 99.90th=[28967], 99.95th=[32900],
    | 99.99th=[34866]
  bw ( KIB/s): min=19466, max=30998, per=99.98%, avg=25011.61, stdev=46.65, samples=1272
  iops : min=4862, max= 7748, avg=6250.34, stdev=11.67, samples=1272
  lat (usec): min=3 228, 250=7.72%, 500=10.69%, 750=4.16%, 1000=0.09%
  lat (msec): 2=0.01%, 4=0.02%, 10=51.72%, 20=21.72%, 50=0.63%
  cpu : usr=0.10%, sys=0.50%, ctx=178942, majf=0, minf=0
  IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
  submit : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  issued rwts: total=26894,62592,0,0 short=0,0,0 dropped=0,0,0
  latency : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=10.5MiB/s (11.0MB/s), 10.5MiB/s-10.5MiB/s (11.0MB/s-11.0MB/s), io=105MiB (110MB), run=10008-10008msec
  WRITE: bw=24.4MiB/s (25.6MB/s), 24.4MiB/s-24.4MiB/s (25.6MB/s-25.6MB/s), io=245MiB (256MB), run=10008-10008msec

Disk stats (read/write):
sda: ios=26624/61963, merge=0/4, ticks=9320/8260, in_queue=17530, util=86.30%
```

运行命令

测试结果说明:

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2. BW, 即磁盘的吞吐量。这个是顺序读写考察的重点。

命令中的常用参数说明：

-filename，指定待测试的文件或块设备的路径，如：-filename=/work/fstest/fio.img(文件)、-filename=/dev/sdb1(块设备)。

-direct，指定当前测试是否采用直接IO方式进行读写，如果采用直接IO则设置为-direct=1，否则设置为-direct=0。建议采用直接IO写测试，这样会使得测试结果更加真实。

-iodepth，IO队列深入，即一次下发的IO的个数，例如：-iodepth=16

-rw，指定读写模式。如：-rw=read(顺序读)、-rw=write(顺序写)、-rw=randread(随机读)、-rw=randwrite(随机写)、-rw=randrw(随机读写，读写的默认比例为5:5)、-rw=randrw-rwmixread=70(指定读写比例为70:30)、-rw=randrw-rwmixwrite=60(指定读写比例为60:40)

-ioengine，指定文件读写方式：sync(采用read、write进行文件读写操作，使用fseek定位读写位置)、psync(采用pread、pwrite进行文件读写操作)、vsync(采用readv、orwritev、进行文件读写操作)。

-bs，指定单次IO的大小；如：-bs=4k

-numjobs，测试进程的并发数，默认为，例：-numjobs=16

-runtime，设置测试运行的时间，单位秒，例：-runtime=300

-group_reporting，关于显示结果的，汇总每个进程的信息

-bssplit，指定设备读写访问的IO大小，如：-bssplit=4k/30 8k/40 16k/30，随机读4k文件占30%、8k占40%、16k占30%

-bsrange，指定IO大小的范围，如：-bsrange=512-2048

-size，指定本次测试文件的大小；如：-size=1G

-sync ，设置同步模式，同步 -sync=1 ，异步 -sync=0

-fsync ，设置数据同步模式，同步 -fsync=1 ，异步 -fsync=0

-lockmem ，对测试所使用的内存进行限制，如： -lockmem=1g ，限制 1G

测试结果说明：

fio的结果报告内容丰富，但我们主要关注下面两项：

1、IOPS，即磁盘的每秒读写次数，这个是随机读写考察的重点。IOPS值越高表示磁盘读写数据越快。IOPS不能设置，是出厂就已经定了，所以IOPS是衡量磁盘读写性能的重要指标之一。

2、BW，即磁盘的吞吐量，这个是顺序读写考察的重点。